

Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

Kaleb K. Javier

June 16, 2026

Strict liability is the primary instrument for environmental harm

- ▶ The firm pays for what it causes, regardless of fault — across oil & gas (OPA 1990), surface mining (SMCRA 1977), and hazardous substances (CERCLA 1980, RCRA 1976)
- ▶ Works when firms can be held fully accountable — breaks down when they cannot (Shavell 1986, 2007)
- ▶ Shavell (2004): two instruments to restore accountability:
 - ▶ Minimum Asset / bonding requirements
 - ▶ **Compulsory liability insurance** — where the premium does the deterrence work, *if* the insurer can price on risk-relevant attributes

This paper examines the design and efficacy of compulsory liability insurance as a financial responsibility instrument in environmental hazard settings.

Does insurance design change firm behavior?

Setting

- ▶ USTs — retail petroleum storage, $\sim$ 140,000 facilities, the most common source of groundwater contamination in the US (EPA 2024)
- ▶ Compulsory insurance the law since 1988 (RCRA Subtitle I, \$1M floor)
- ▶ **Clean shock:** Texas exits its flat-fee fund December 22, 1998 — 26,000 facilities enter risk-rated private insurance overnight
- ▶ 9.8M tank-years of EPA panel data spanning both regimes

Three questions

- ▶ **Q1 — Observability:** Is release risk predictable from observable tank characteristics?
- ▶ **Q2 — Behavior:** Does risk-based pricing change closure, replacement, and release outcomes?
- ▶ **Q3 — Welfare:** How much of the first-best welfare gain does risk-rated pricing capture?

What is an underground storage tank?

[ADD FIGURE HERE]

Related literature

Liability regimes, financial responsibility, and UST policy

Shavell (1986, 2004, 2007); Yin, Kunreuther & White (2011); Yin, Pfaff & Kunreuther (2011, 2007); Boomhower (2019); Ho, Hsu, Cha & Rivera (2018)

First firm-level panel analysis of insurance design — risk-rated vs. flat-fee — as a financial responsibility instrument; clean policy shock; structural welfare decomposition

Environmental economics and environmental risk

Fowlie, Reguant & Ryan (2016); Muehlenbachs (2015); Blundell, Gowrisankaran & Langer (2020); Kellogg & Reguant (2021)

Dynamic structural model of firm behavior under environmental regulation; welfare decomposition bounding gap to first-best

IO of selection markets: risk-based and cost-based pricing

Einav, Finkelstein & Mahoney (2021); Einav, Finkelstein & Cullen (2010); Bundorf, Levin & Mahoney (2012); Handel, Hendel & Whinston (2015); Mahoney & Weyl (2017)

[PLACEHOLDER]

Setting, Data, and Observability

Who operates USTs?

Operator type	Share of facilities	Share of tanks
Independent retail (gas station / c-store)	[XX]%	[XX]%
Branded / major oil company	[XX]%	[XX]%
Fleet / non-retail (trucking, municipal)	[XX]%	[XX]%
Government / other	[XX]%	[XX]%

- ▶ **[xyz type is dominant type in sample]** — retail gas stations, truck stops, and fleet facilities account for [XX]% of operators; branded majors are a minority
- ▶ ~,140,000 active facilities at peak; ~,550,000 tanks nationwide
- ▶ Median facility: [X] active tanks, [XXX]k gallons total capacity

Release risk varies with observable tank traits

- ▶ Annual release probability is driven by **tank age** and **wall construction**
- ▶ Single-walled steel: no secondary containment, corrodes gradually, hazard rises steeply with age
- ▶ Double-walled: interstitial monitoring space, hazard flat and low across all ages
- ▶ Key observables: age, wall type, piping, leak-detection technology — exactly what an underwriter can verify

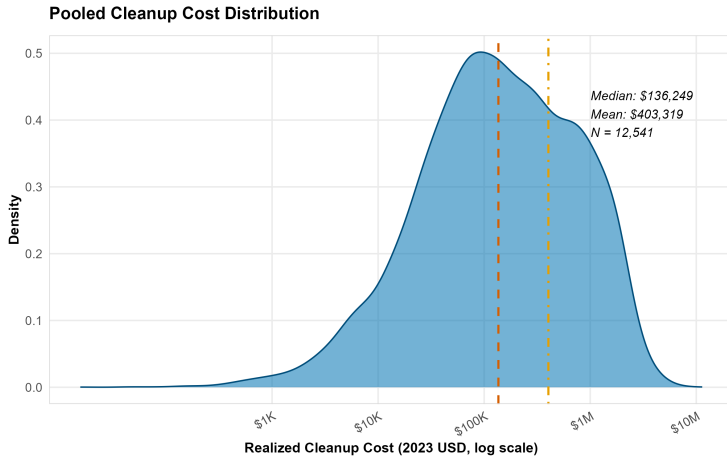
Whether the private market prices this heterogeneity is an empirical question — answered after we set up the data.

The harm is silent, slow, and diffuse — tort law cannot reach it

- ▶ No secondary containment, no visible event, no immediate victim
- ▶ Benzene and petroleum migrate silently into groundwater; by detection the plume may have traveled hundreds of feet
- ▶ Diffuse victims, delayed injury, causation hard to establish — private tort law does not operate here
- ▶ Health damages ~,\$17k per release (Marcus 2021) generate no insurance claim: contamination unobservable, victims cannot sue

The regulatory system — compulsory insurance — is the only operative instrument.

Releases are costly to cleanup budgets and human health



~ 550,000 confirmed releases since 1988. Median cleanup \$136k; mean \$403k. Right tail dominates expected loss.

The choice set is effectively binary: operate or exit

Action	Risk effect	Why it is closed off
Spill / corrosion protection	Partial reduction	Mandated by 1998 EPA deadline; near-universal by reform
Leak-detection up-grade	Monitoring only	Mandated by 1993 EPA deadline; near-universal by reform
Partial wall up-grade	Could reduce risk	Triggers full facility reclassification under 40 CFR Part 280 — cost dwarfs any premium saving
Replace with new DW tank	Large reduction	~\$55k/tank; viable but rare
Remove / close	Eliminates hazard	The operative margin

Federal compliance deadlines (1988–1998) mandated spill protection, corrosion control, and leak detection. By the time of the reform, these margins were largely exhausted. Only **0.003%** of active Texas facilities ever retrofitted tank construction prior to terminal closure, 1998–2021. **Removal is the only lever.**

Financial responsibility: three approaches in practice

RCRA Subtitle I (1988) requires all UST operators to demonstrate financial capacity to cover a confirmed release — a federal \$1M floor. Three approaches exist:

Approach	Examples	Price signal to operator
No state fund (private market only)	OR, WA	Depends on private insurer; no co-ordinated premium mandate
Flat-fee state fund (pooled trust)	38 states, incl. TX pre-1998	None: flat per-tank fee regardless of age, wall type, or leak history $\Rightarrow \partial P / \partial \text{age} = 0$
Risk-rated private insurance (compulsory)	TX post-Dec 1998	Strong: premium rises with age, wall construction, piping $\Rightarrow \partial P / \partial \text{age} > 0$

Texas exits its flat-fee fund December 22, 1998 $\rightarrow$ 26,000 facilities enter risk-rated private insurance overnight.

Data: three collection efforts, 18 usable states

- ▶ **State + EPA administrative records** — UST inventory and confirmed releases from all 50 states and EPA LUST Finder; tank-level attributes (install date, wall type, piping, leak detection) and cleanup dispositions
- ▶ **Texas record requests** — TX UST registry; FR insurance contract data and carrier-level coverage composition, 1993–2021
- ▶ **TX insurance rate filings** — Mid-Continent Casualty SERFF filings, 2006–2024; reconstructs the risk-based premium schedule $P(s)$ at the tank level from base rates, ILFs, and attribute credits/debits

Why 18 states? 20 of 38 flat-fee states dropped: missing closure dates, install dates, or wall construction variables. The 17 controls plus Texas have complete records across all three dimensions.

	Full sample	Texas	Controls
<i>Coverage (alive-at-reform sample)</i>			
Facilities	252,670	54,379	198,291
Tanks at reform	330,311	60,651	269,660
Facility-years	4.82M	0.90M	3.93M
<i>At reform (Dec 22, 1998)</i>			
Share single-walled	89%	89%	89%
Share pre-1989 vintage	69%	75%	68%
Median tank age (years)	14.0	14.4	14.0
<i>Average facility profile (full sample)</i>			
Tanks per facility (med. [IQR])	2 [1–3]	3 [2–3]	2 [1–3]
Total capacity, k gal (med. [IQR])	15 [4–27]	20 [8–30]	14 [3–26]
Mean tank age (years)	18.8	15.5	19.6
Annual release rate	1.4%	0.5%	1.6%

18 study states; alive-at-reform = facilities with ≥ 1 tank active on Dec 22, 1998. Wall, vintage, and age computed over tanks active at reform; profile averages over all study facility-years. Capacity is the median (robust to large-facility outliers).

Risk is predictable: the age-hazard gradient (setup)

Estimand. Annual first-release hazard in the never-yet-leaked risk set:

$$h(t \mid x) = P(\text{first confirmed release in year } t \mid \text{no prior release, } x)$$

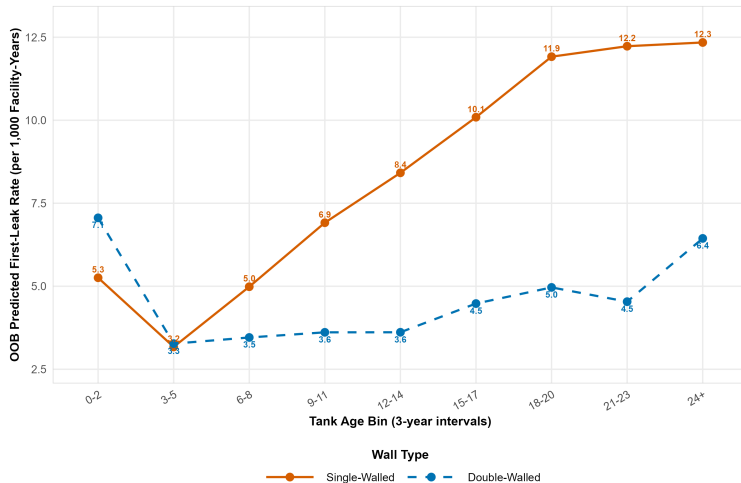
In plain words: given what an underwriter can observe at the start of a policy year, what's the probability of a claim?

Model. Penalized logistic regression on the pre-treatment panel:

- ▶ x : 3-year age bins fully interacted with wall construction; state FE; fuel and portfolio controls
- ▶ Lasso variable selection; inverse-frequency weights for the 0.70% base rate
- ▶ Out-of-sample CV (10 folds); post-hoc recalibration to the empirical event rate

Sample. 3.93M pre-treatment facility-years; 27,455 confirmed first-release events. $\hat{h}(x)$ feeds the next figure and the structural-model hazard $h(a_t, W_i)$.

Single-walled tanks drive the age-hazard gradient



The insurance market: how Texas premiums are built

Texas — rebuilt from major Texas UST Insurer's rate filings (SERFF, 2006–2024):

$$P_{ijt} = \text{Base}_t \times \text{ILF}_{it} \times \left(1 + \sum_k \lambda_{k,t} X_{ijt,k} \right)$$

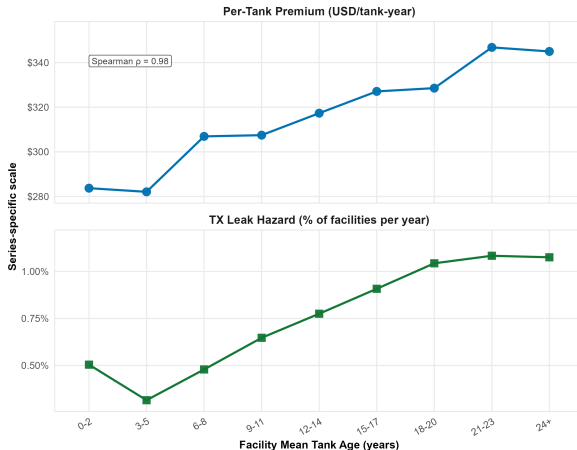
- ▶ Base_t : per-tank base rate, four filing periods
- ▶ ILF_{it} : increased-limit factor; fix coverage at federal-minimum \$1M/\$1M
- ▶ X_{ijt} : tank age, wall, piping, leak-detection technology
- ▶ $\lambda_{k,t}$: additive credits/debits from the filed schedule

Control states — state financial-responsibility fund records: a per-tank annual assessment, largely flat across years and tank characteristics.

Premiums track the risk we can predict

Premium Schedule Tracks Texas-Estimated Leak Hazard

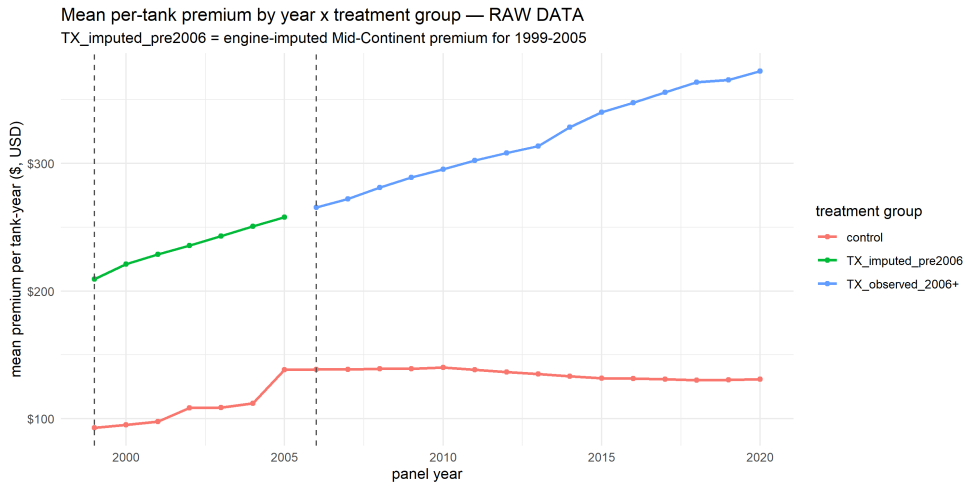
Per-tank premium (TX Mid-Continent, 2006-2014) and TX leak hazard (01n elastic net OOB) both rise across age bins



Notes: Per-tank premium = mean(tank_premium) across TX Mid-Continent tank-months in each bin. TX leak hazard = mean of row-level

Predicted per-tank EL →

Premium divergence: Texas vs. control states (1992–2020)



Flat and parallel before 1998; sharp Texas divergence after the reform. Sets up the DiD without showing it

Reduced Form Evidence

Research design: DiD with facility FE and cell-by-year FE

Static DiD.

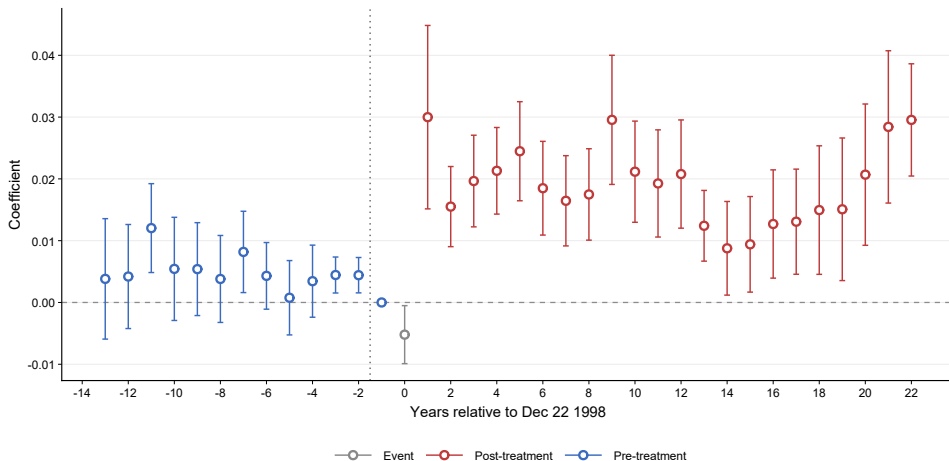
$$Y_{it} = \beta (\text{Texas}_{f(i)} \times \text{Post}_t) + \alpha_i + \delta_{ct} + \varepsilon_{it}$$

Event study.

$$Y_{it} = \sum_{\tau \neq -1} \beta_{\tau} \cdot (\text{Texas}_{f(i)} \times \mathbf{1}[t - 1998 = \tau]) + \alpha_i + \delta_{ct} + \varepsilon_{it}$$

- ▶ α_i : facility FE. δ_{ct} : cell $\times$ year FE with $c = \{\text{wall} \times \text{fuel} \times \text{capacity} \times \text{install cohort}\}$
- ▶ *Identifying variation*: within a single tank-vintage cell, in the same calendar year, β is the change in the Texas-vs-control closure-rate gap before vs. after Dec 22, 1998

Event study: closure response in Texas



Pre-trends flat; sharp break at Dec 22, 1998; sustained post-treatment effect.

ATT on Tank Closure: +1.58 pp on a 2.0% Baseline

	(1) Raw (1)	(2) + Fac FE (2)	(3) + Controls (3)	(4) + Cell FE (4)
Constant	0.0203*** (0.0022)			
did_term	0.0106*** (0.0022)	0.0338*** (1×10^{-6})	0.0350*** (0.0015)	0.0158*** (0.0040)
mandate_release_det			-0.0039 (0.0035)	
mandate_spill_overfill			0.0048* (0.0025)	
mandate_integrity			0.0085*** (0.0026)	
Observations	9,761,982	9,761,982	9,761,982	9,761,044
R ²	0.00047	0.03550	0.03584	0.07745
Within R ²		0.00218	0.00253	0.00036
panel_id fixed effects		✓	✓	✓
cell_vintage_year_fe fixed effects				✓

Cols add facility FE, then cell×year FE. Col (3) is the headline.

Heterogeneity: Pre-89 Vintage Drives the Effect

	(1) Raw (1)	(2) + Fac FE (2)	(3) + Controls (3)	(4) + Cell FE (4)
Constant	0.0103*** (0.0011)			
did_term	0.0048*** (0.0011)	0.0117*** (0.0028)	0.0114*** (0.0029)	-0.0067** (0.0031)
pre89_cohort	0.0135*** (0.0023)	0.0671*** (0.0062)	0.0696*** (0.0060)	
did_term × pre89_cohort	0.0094*** (0.0023)	0.0382*** (0.0029)	0.0360*** (0.0029)	0.0309*** (0.0065)
mandate_release_det			-0.0114** (0.0040)	
mandate_spill_overfill			0.0009 (0.0028)	
mandate_integrity			0.0044 (0.0030)	
Observations	9,761,982	9,761,982	9,761,982	9,761,044
R ²	0.00249	0.04685	0.04750	0.07772
Within R ²		0.01392	0.01460	0.00065
panel_id fixed effects		✓	✓	✓
cell_vintage_year_fe fixed effects				✓

$$\text{Pre-89 ATT} = \hat{\beta}_0 + \hat{\beta}_1 \approx +3.1 \text{ pp}; \text{ post-88 ATT} = \hat{\beta}_0 \approx +0.6 \text{ pp}.$$

LUST discovery falls under risk-based pricing

Outcome	Pre-reform ctrl mean	TX $\times$ Post
Any leak found this year	1.73%	-1.27 pp ^{***} (0.0032)
<i>By how the leak was discovered:</i>		
Routine monitoring (no closure in $\pm 60d$)	1.16%	-0.76 pp ^{***} (0.0021)
Closure-inspection finding (closure within 60d)	0.61%	-0.51 pp ^{**} (0.0017)
N facility-years	4,650,240	
Sample	Incumbent (alive at Dec 22, 1998)	
FE	Facility + cell $\times$ year; cluster by state	

Leak discovery falls 1.27 pp on a 1.73% base; both routine-monitoring and closure-inspection findings decline.

Closure composition shifts toward replacement

Outcome (conditional on closing)	Pre-reform ctrl mean	TX × Post
Facility closes all tanks this year	70.9%	+6.5 pp ^{***} (0.019)
Has a permanent closure (no replace)	88.3%	−6.2 pp ^{***} (0.012)
Has a replacement closure (upgrade)	12.0%	+6.3 pp ^{***} (0.013)
Any DW tank installed	4.1%	−1.4 pp [*] (0.007)
Net tank change (count)	−1.905	+0.020 (0.085)
Capacity change (gal)	−100,394	−1,397 (5,715)
<i>N</i> closing facility-years	59,107	
Sample	Incumbent, any_closure = 1	
FE	Facility + make_model_fac × year; cluster by state	

Among closing facilities, RBP raises the replacement (upgrade) share and lowers permanent closures — the margin the structural model recovers.

What the reduced form cannot tell us

The DiD identifies an average treatment effect. Three questions it cannot answer:

1. **How large are the welfare effects in dollar terms?** The DiD gives us a closure-rate gap; it does not give us a present-value welfare number on which to compare policies.
2. **What would alternative policies — subsidy, Pigouvian surcharge, age mandate — actually do?** These policies were not implemented in Texas. Their welfare consequences are counterfactual.
3. **How far are we from first-best, and why?** RBP is a partial fix: premiums price what the insurer can claim, but health damages are structurally unpriced. Quantifying the residual wedge requires the structural primitives.

Recovering the structural parameters ($\kappa, K, \gamma_p, \gamma_r$) is the only way to answer these. The structural model is not an add-on — it is required.

Structural Model and Counterfactuals

A dynamic discrete-choice model of tank operations

- ▶ Each period, a facility chooses one of three actions: **Maintain**, **Exit**, or **Replace**
- ▶ The choice trades off current flow payoffs against the continuation value of future operation
- ▶ Framework: optimal stopping (Rust 1987); estimated by Nested Pseudo-Likelihood (Aguirregabiria & Mira 2007)

Flow utilities: one payoff per action

$$u^M(s, g) = \underbrace{1}_{\text{net rev.}} + \gamma_{\text{price}} P(s) - \gamma_{\text{risk}} h(s) + \alpha_g \quad (\text{maintain})$$

$$u^E(s) = \kappa_w(s) \quad (\text{exit / scrap})$$

$$u^R(s, g) = -K_{w(s)} + \alpha_g \quad (\text{replace})$$

- ▶ $P(s)$ premium (SERFF filings), $h(s)$ leak hazard (risk model) — both vary by (age, wall, regime)
- ▶ **Estimated** (in green): κ_w scrap value, K_w replacement cost, γ_{price} price sensitivity, γ_{risk} risk internalization
- ▶ α_g state $\times$ regime stay FE — a measurement correction, profiled out; it does not enter the counterfactual

Net revenue R normalized to \$10k/yr/facility; all values in this scale.

Keep, scrap, or replace: the dynamic stopping problem

$$V(s) = \max \left\{ \underbrace{u^M(s) + \beta \mathbb{E}[V(s') \mid s]}_{\text{Maintain}}, \underbrace{\kappa_{W(s)}}_{\text{Exit}}, \underbrace{-K_{W(s)} + \beta \mathbb{E}[V(s_{\text{new}})]}_{\text{Replace}} \right\}$$

- ▶ **Maintain** — flow payoff today, keep the option to operate next year
- ▶ **Exit** — take κ_W and leave (absorbing)
- ▶ **Replace** — pay K_W , restart with a new DW tank at age 0

Regime. Flat fees: premium flat in age $\Rightarrow$ aging carries no price penalty. Risk-based: premium rises with age $\Rightarrow$ continuation value falls $\Rightarrow$ firms exit/replace sooner.

Stopping rule: close once scrap value exceeds the value of continuing; RBP lowers the age at which that binds.

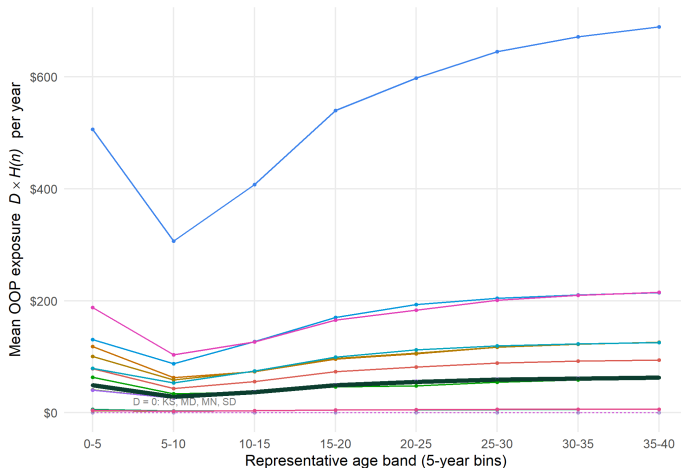
The same tank in two worlds

A thought experiment. Take one tank — 28 years old, single-walled, same leak risk in either world — and drop it into each pricing regime:

	Flat-fee world (control state)	Risk-based world (Texas)
Annual premium	\$130 (flat)	\$380 (rises with age)
Leak risk $h(s)$	high	high (<i>same</i>)
$\Rightarrow$ closes the tank	sometimes	much more often

- ▶ It already closes *sometimes* under flat fees — that response to **risk alone** is γ_{risk}
- ▶ The **extra** closing once the premium jumps, risk unchanged, is γ_{price}
Same tank, same risk — only the premium differs. The within-regime response gives risk; the across-regime gap gives price.

$D \times \text{hazard}$: the variation that identifies the γ 's



Flat-fee cells (orange) sit at low deductibles across the hazard range; risk-based cells (teal) price the hazard. Deductible variation at given hazard pins γ_{price} ; the hazard spread pins γ_{risk} .

What identifies each parameter

Parameter	Moment in the data
κ_{SW}, κ_{DW}	Exit rate by wall type — higher scrap value $\Rightarrow$ more exit
K_{SW}, K_{DW}	Replace-vs-exit share among closing facilities, by wall
γ_{price}	Choice-share response to premium variation across cells
γ_{risk}	Residual age-exit gradient once premium is accounted for
α_g	Group-level stay rate (profiled out — measurement only)

The cross-regime contrast (FF: only hazard varies; RB: both vary) is what separates γ_{price} from γ_{risk} .

κ exit share $\rightarrow$

K replace share $\rightarrow$

Structural estimating sample

Unit of observation. Facility-year. One row per facility per year, 1999–2018.

Sample. Texas 2006+ (post-RB implementation) plus 17 control states 1999+. The structural model fits the *equilibrium* of the risk-based regime — TX is the regime that's actually priced.

- ▶ $N \approx 2.28\text{M}$ facility-years after data cleaning
- ▶ Choices observed: maintain / exit / replace; mapped onto 32 state cells (a, W, R)
- ▶ Premiums p_t from Mid-Continent rate-filing reconstruction; hazards $h(a_t)$ from the first-release-prediction model

Parameter estimates: 6-parameter NPL with profiled state FE

	Estimate
$\hat{\kappa}_{SW}$ scrap value, single-walled	\$252,349 (\$1,213)
$\hat{\kappa}_{DW}$ scrap value, double-walled	\$243,800 (\$1,239)
$\hat{K}_{SW}$ replacement cost, single-walled	\$51,383 (\$176)
$\hat{K}_{DW}$ replacement cost, double-walled	\$51,162 (\$372)
$\hat{\gamma}_{price}$ price sensitivity	-2.163 (0.205)
$\hat{\gamma}_{risk}$ risk internalization	+0.114 (0.005)
Log-likelihood	-227,240
Facility-years	2,282,735
State stay fixed effects	18

Standard errors in parentheses (Aguirregabiria-Mira 2002 profile likelihood). κ and K in dollars; K standard errors via the delta method on the log scale. Both γ are highly significant ($t = -10.6$ and 21.6).

What the estimates say

- ▶ $\hat{\kappa}_{SW} \approx \$252\mathbf{k}$, $\hat{\kappa}_{DW} \approx \$244\mathbf{k}$. Scrap value on exit — close across wall types after the cleanup.
- ▶ $\hat{K}_{SW} \approx \hat{K}_{DW} \approx \$51\mathbf{k}$. New-tank cost barely moves with wall type — wall matters through hazard and premium, not installation.
- ▶ $\hat{\gamma}_{\text{price}} = -2.16$. Firms respond strongly to the premium signal: higher premiums push toward exit and replacement.
- ▶ $\hat{\gamma}_{\text{risk}} = +0.114$, **small**. Firms act on only a sliver of expected leak damage — most is externalized. **That wedge is what the counterfactuals quantify.**

Each policy changes one term in the payoff

Baseline (risk-based): $u^M(s) = 1 + \gamma_{\text{price}} P(s) - \gamma_{\text{risk}} h(s) + \alpha_g$

CF1 — Flat-fee: $u^M(s) = 1 + \gamma_{\text{price}} \bar{P} - \gamma_{\text{risk}} h(s)$

CF2 — Replacement subsidy: $u^R(s) = - (1 - s^*) K_w, \quad s^* = 0.50$

CF3 — Pigouvian: $u^M(s) = 1 + \gamma_{\text{price}} P(s) - \gamma_{\text{risk}} h(s) - h(s) E$

CF4 — Age mandate: $u^M(s) = -\infty$;, for tanks 25+ years old

*Each policy re-solves V and the stationary fleet; only the **highlighted** term moves.*

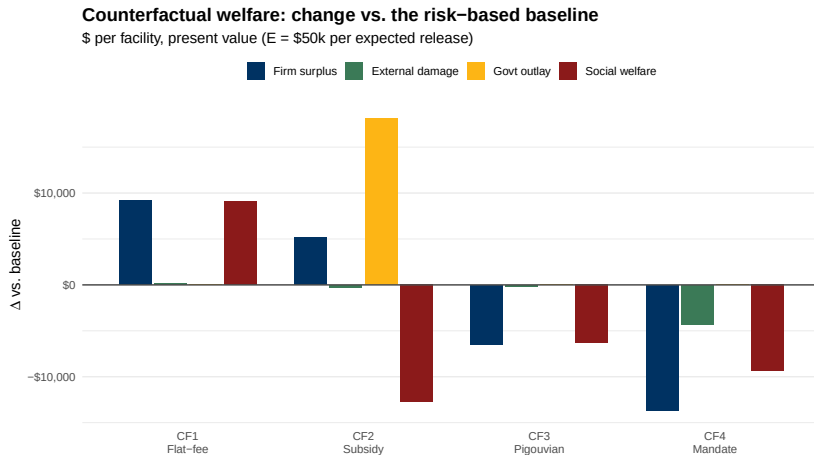
Welfare accounting

$$\Delta W = \underbrace{\sum_s \pi(s) \Delta V(s)}_{\Delta \text{ Firm surplus}} - \underbrace{\sum_s \pi(s) \sum_t \beta^t h(s_t) (L + E)}_{\Delta \text{ External damage}} - \underbrace{\Delta \text{Govt outlay}}_{\text{subsidy only}}$$

- ▶ **Firm surplus** — change in the discounted value to operators
- ▶ **External damage** — unpriced cleanup + health harm, $L + E$ per expected release
- ▶ **Govt outlay** — subsidy cost (CF2 only)

$E = \$17k$ (health) or $\$50k$ (with unmeasured damages). State fixed effects do not enter the counterfactual re-solve.

Counterfactual welfare: change in firm surplus, external damage, a



Change in each welfare component relative to the risk-based baseline, by counterfactual

Counterfactual welfare summary

Counterfactual	Δ Firm surplus	Δ External damage	Δ Govt outlay	Δ Social welfare
CF1 — Flat-fee	+9,213	+120	0	+9,092
CF2 — Replace subsidy	+5,148	-261	+18,170	-12,761
CF3 — Pigouvian	-6,521	-194	0	-6,327
CF4 — Age mandate	-13,686	-4,297	0	-9,389

Δ vs. the risk-based baseline; \$/facility present value, $E = \$50k$. Positive = improvement over the baseline.

Takeaways

1. **Risk is observable.** The age-hazard gradient is steep for single-walled tanks and flat for double-walled. Shavell's observability condition is satisfied in this market.
2. **Risk-based pricing shifts behavior.** +1.58 pp annual closure ATT; $\sim 4\times$ larger in the pre-1989 SW-heavy stock that the premium schedule actually prices. Composition shifts toward the replacement margin (12% $\rightarrow$ 18%). LUST discovery falls through both background and inspection-triggered channels.
3. **Welfare gap to first-best persists.** [PLACEHOLDER: directional claim about the four counterfactuals at $E = \$17k$, pending 6p re-run. Even after the reform, the health-externality component remains structurally unpriced — actuarial premiums cannot price what no claimant recovers.]

Appendix: The Agent's Decision (toy model)

The Choice. Each period t , the facility chooses $d_t \in \{\text{Maintain}, \text{Close}\}$.

The Trade-off.

- ▶ **Maintain:** Earn net revenue, pay insurance premiums.
- ▶ **Close:** Exit and recover scrap value κ (liquidity of land).

Flow Utility (Maintain).

$$u(a) = \underbrace{R}_{\text{Net Revenue}} - \underbrace{C_j(a)}_{\text{Insurance Premium}}$$

Simplifying assumption (full coverage): environmental risk enters as an *ex-ante* premium cost, not an *ex-post* shock.

Two-action stopping model used in early presentations — included for completeness; superseded by the three-action DCC in the main deck.

Appendix: The Dynamic Problem (toy model)

Bellman equation.

$$V(a) = \max \left\{ \underbrace{u(a) + \beta \mathbb{E}[V(a') \mid a]}_{\text{value of maintaining}}, \underbrace{\kappa}_{\text{scrap value}} \right\}$$

Continuation value. The value of keeping the tank active today to preserve the option of operating tomorrow:

$$V_{\text{cont}}(a) = u(a) + \beta \mathbb{E}[V(a') \mid a]$$

Optimal stopping rule. Close if $V_{\text{cont}}(a) < \kappa$.

Appendix: The Policy Lever: Insurance Regimes (toy model)

1. Uniform Premium Pooling (F) — 38 states.

- ▶ Premium: constant $\bar{P}$, regardless of age or risk.
- ▶ Marginal cost of aging: $\partial C / \partial a = 0$. No incentive to close aging tanks.

2. Risk-Based Pricing (RB) — Texas post-1999.

- ▶ Premium: rises with the leak hazard, $C(a) = \lambda \cdot h(a) \cdot L$.
- ▶ Marginal cost of aging: $\partial C / \partial a > 0$. Aging tanks become increasingly costly.

Appendix: What's the First Best? (toy model)

Social planner's objective. Internalize remediation costs (L) and health externalities (E):

$$u_{\text{SOC}}(a) = R - h(a) \cdot \left(\underbrace{L}_{\text{Remediation}} + \underbrace{E}_{\text{Health}} \right)$$

Why risk-based pricing falls short.

- ▶ Actuarial premiums reflect insurer claims data $\rightarrow$ only remediation costs (L)
- ▶ Health damages (E) generate no third-party claims: contamination is unobservable (Marcus, 2021)
- ▶ $\Rightarrow$ Even fair risk-based pricing leaves E externalized

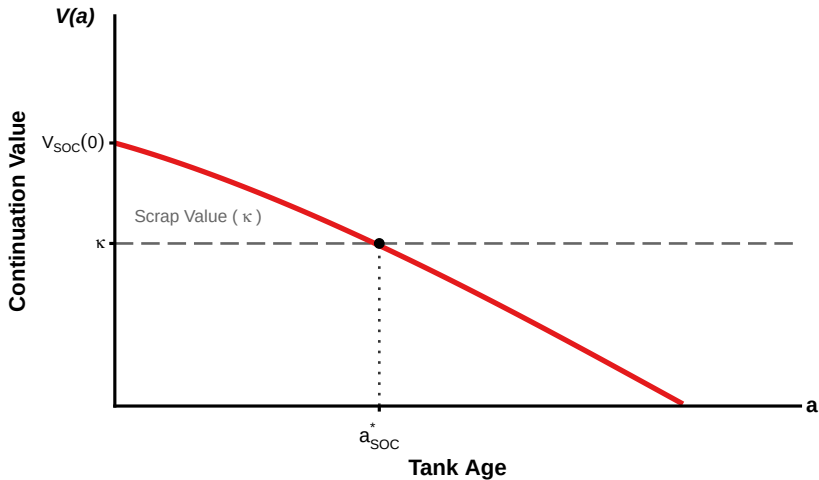
Implication. $V_{\text{RB}}(a) > V_{\text{SOC}}(a)$ — RBP induces *some* closures, but not enough.

Appendix: Three Facts to Keep in Mind (toy model)

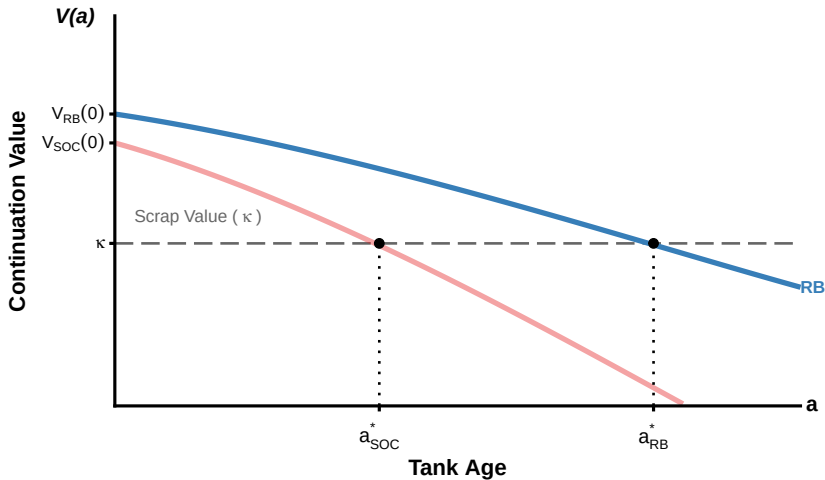
- 1. Asset depreciation.** Rising leak risk drives tank value strictly down in age ($V'(a) < 0$). The regime determines whether private incentives align with this social decay.
- 2. Inefficient retention (Uniform Premium).** Static costs $\bar{P}$ imply zero marginal cost of aging. Firms ignore risk and retain assets longest: a_{UP}^* .
- 3. Partial correction (Risk-Based).** Rising premiums $h(a)L$ imply positive marginal cost of aging. Closure accelerates to a_{RB}^* , but still $> a_{SOC}^*$ due to unpriced health damages.

$$a_{SOC}^* < a_{RB}^* < a_{UP}^*$$

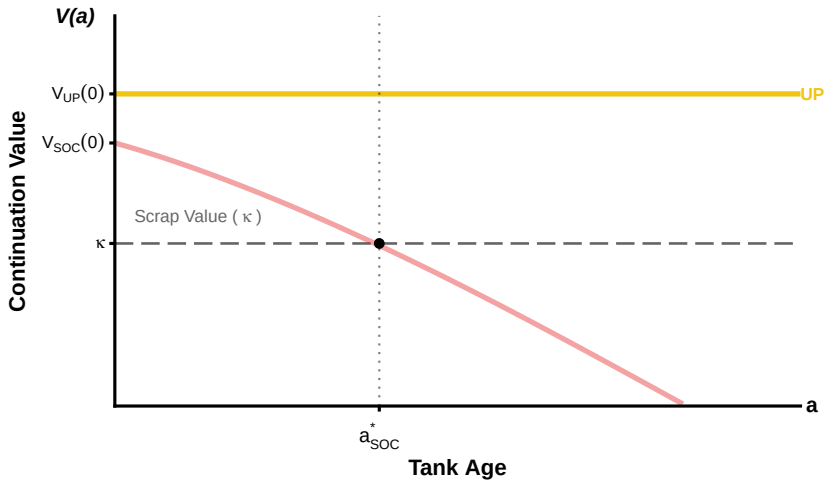
Appendix: Social Optimum Exit Threshold (toy model)



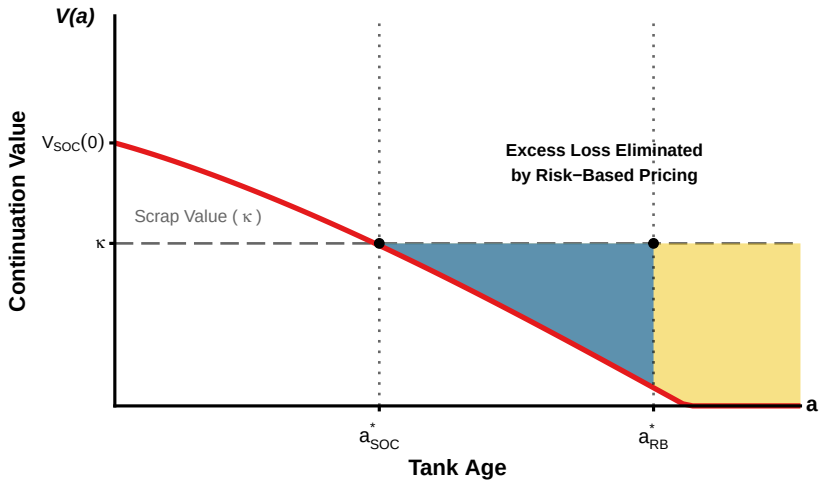
Appendix: Risk-Based vs. Social Optimum (toy model)



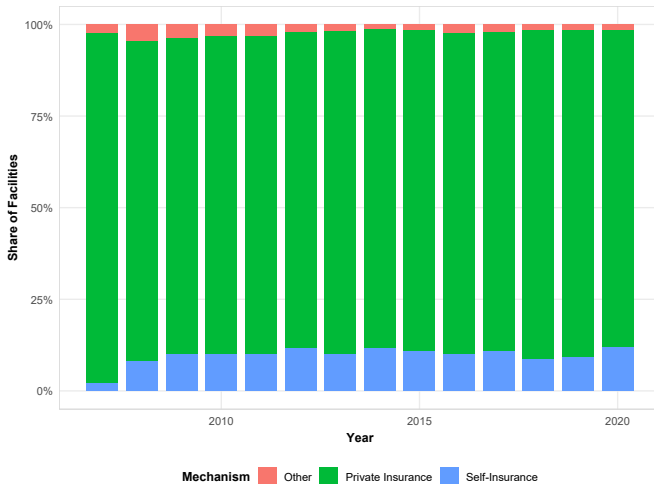
Appendix: Uniform Premium vs. Social Optimum (toy model)



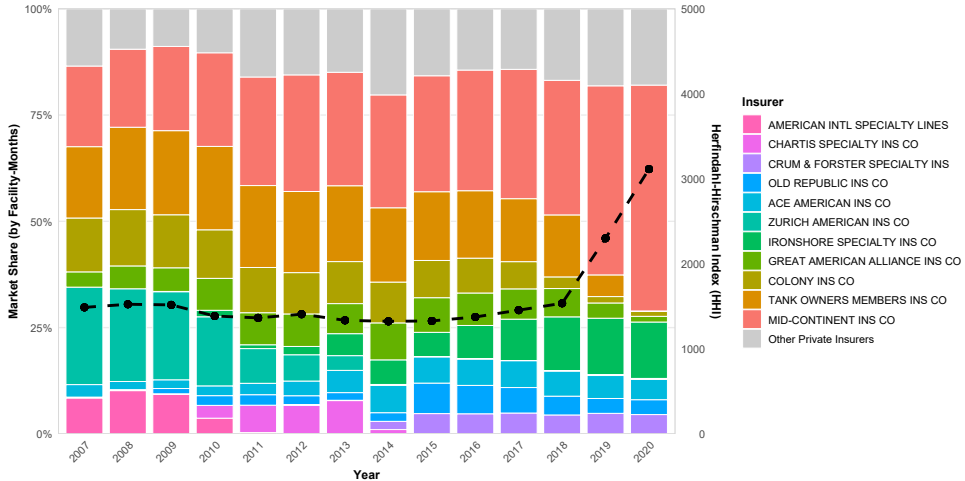
Appendix: Deadweight Loss Across Regimes (toy model)



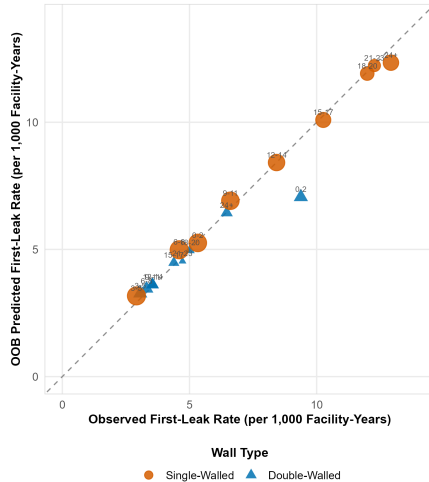
Appendix: FR Compliance: TX is a Private-Insurance Market



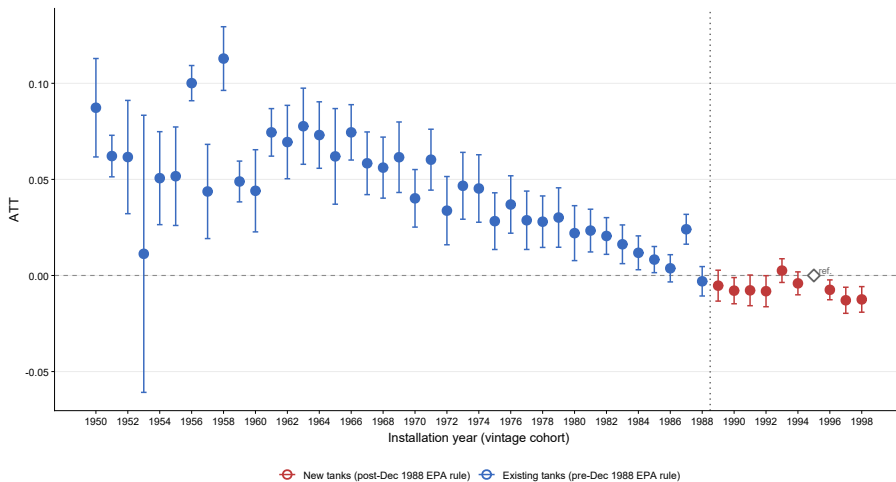
Appendix: Market Structure: Concentrated and Stable



Appendix: Cell-Level Goodness of Fit



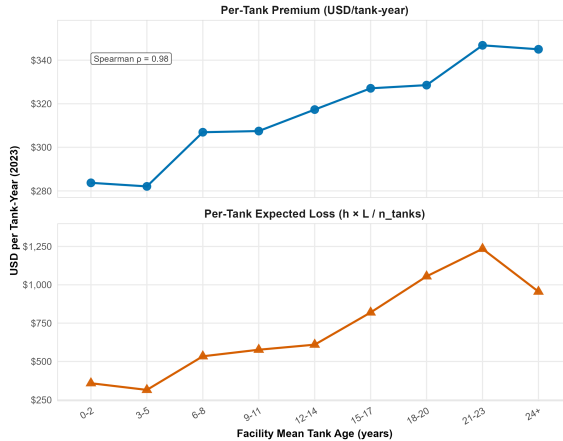
Appendix: ATT by Installation Vintage



Appendix: Premium vs. Per-Tank Expected Loss

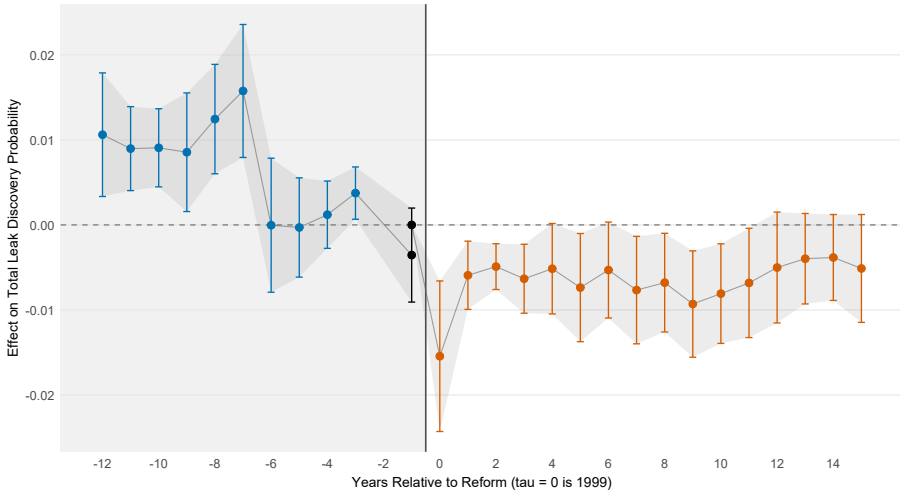
Premium and Constructed Per-Tank Expected Loss Across Age Bins

Backup view: per-tank EL = $\hat{h} \times \hat{L} / n_{\text{tanks}}$, where $\hat{L}$ is a control-state donor curve (no TX cost data exi

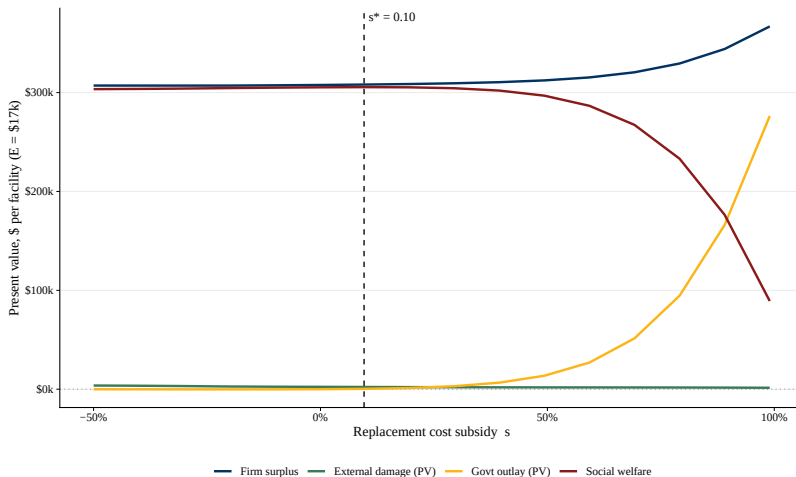


Notes: Per-tank premium as in main figure. Per-tank expected loss = $\hat{h} \times \hat{L} / n_{\text{tanks}}$, computed at the tank-month level using ea

Appendix: LUST Event Study — Total Leak Discovery



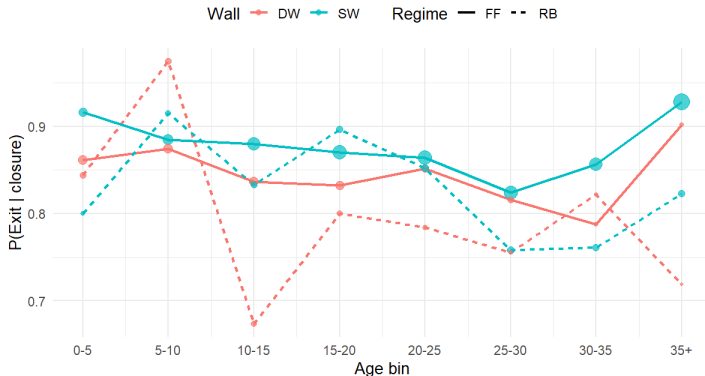
Appendix: Subsidy Curve



Appendix: Exit share among closures, by age and wall

Exit share among closures, by age and wall

Level and age-profile identify kappa_SW, kappa_DW

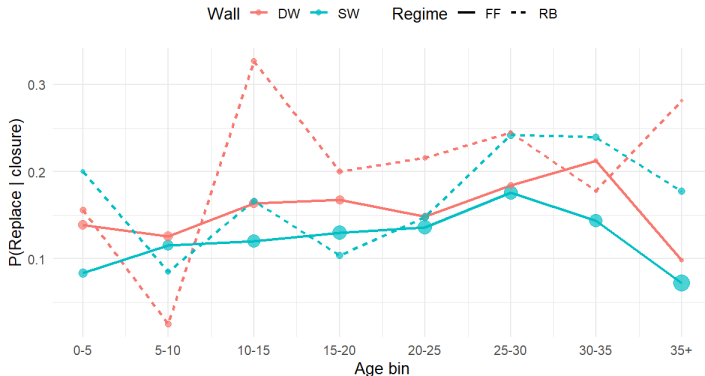


The level and age-profile of full exit identify the scrap values κ_{SW} , κ_{DW} .

Appendix: Conditional replace share among closures

Conditional replace share among closures

Both walls carry exit-vs-replace variation -> identifies K

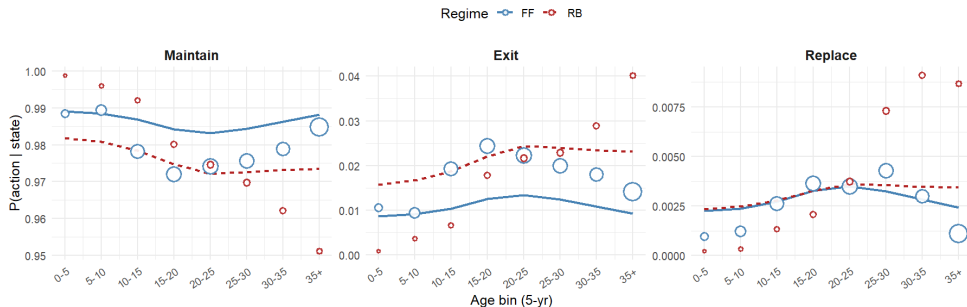


Both wall types show genuine exit-vs-replace variation, which identifies K_{SW} and K_{DW} .

Appendix: Model fit — SW panel

6p+FE fit — Single-Walled (SW) (T007 Phase 3, observed sample)

Solid/dashed line = MODEL-IMPLIED $P(\text{action} \mid \text{state})$; open circle = EMPIRICAL share (size $\propto n$ obs)



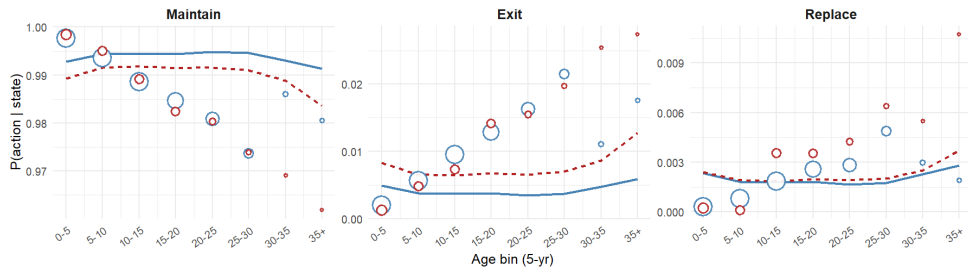
Solid lines: model-implied CCPs; open circles: empirical cell shares. 32 cells fit simultaneously across three actions; SW panel.

Appendix: Model fit — DW panel

6p+FE fit — Double-Walled (DW) (T007 Phase 3, observed sample)

Solid/dashed line = MODEL-IMPLIED $P(\text{action} \mid \text{state})$; open circle = EMPIRICAL share (size $\propto n$ obs)

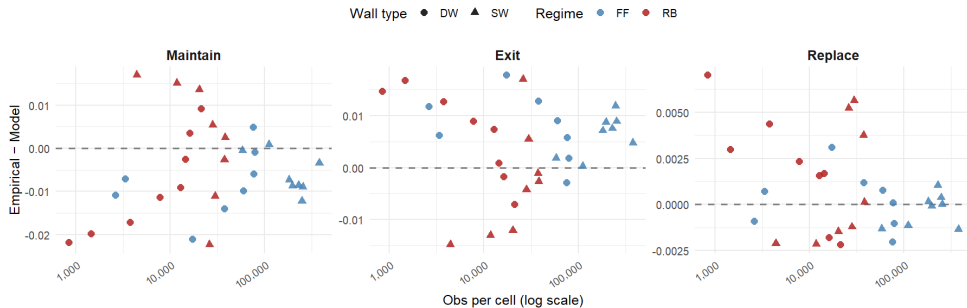
Regime — FF — RB



Appendix: Per-cell fit residuals (6p)

6p+FE per-cell residuals: empirical share – model-predicted P (T007 Phase 3)

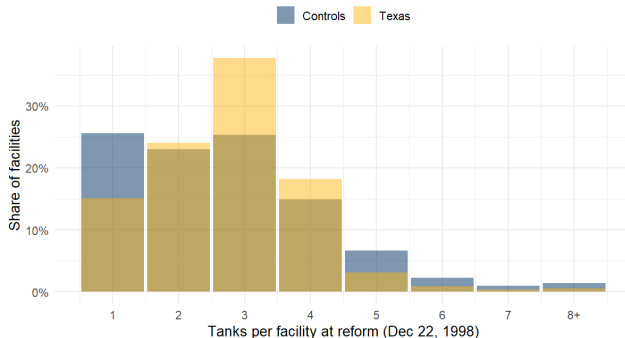
Dashed = perfect fit; x-axis = obs per cell (log scale)



Appendix: Facility size at reform — Texas vs. controls

Facility size at reform: Texas vs. controls

Tank count per incumbent facility (overlapping)

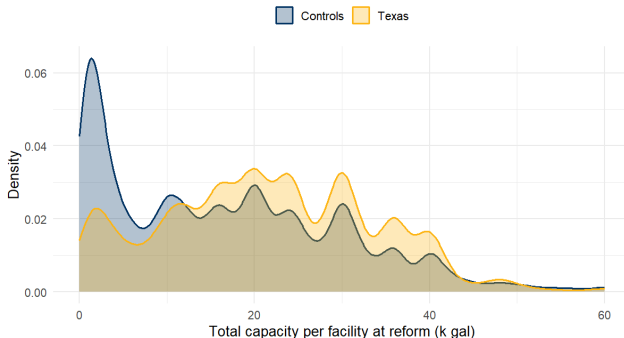


Tank count per incumbent facility on Dec 22, 1998. Texas facilities skew larger (more 3–4 tank facilities); controls have more single-tank facilities.

Appendix: Facility capacity at reform — Texas vs. controls

Facility capacity at reform: Texas vs. controls

Total tank capacity per incumbent facility, Dec 22 1998 (clipped at 60k gal)



Total tank capacity per incumbent facility on Dec 22, 1998 (clipped at 60k gal). Controls have a large mass of small single-tank facilities; Texas capacity is shifted right.

Appendix: What the model captures — and where it strains

The model. Single-agent optimal-stopping DDC. Each *facility* = one representative tank on a 32-cell grid (age $\times$ wall $\times$ regime); each year it chooses **maintain / exit / replace**. **Captures well.** The dominant margin — **exit** as tanks age, and its response to the premium. Full exits are coded correctly; γ_{price} and γ_{risk} are identified off this well-measured margin. **Strains** — all from one assumption, *a facility is a portfolio, not one tank*:

- ▶ **No partial-closure action** — downsizing (close one tank, keep operating) is coded Maintain. Small overall (0.36%), concentrated at large facilities.
- ▶ **Mis-specified “replace” transition** — real upgrades are partial, so 98% land off the model’s reset-to-one-new-tank support.
- ▶ **Omitted state (facility size)** — choice probabilities vary by size; the model averages over it.

Clean: maintain / exit / premium. Strained: the portfolio (size + downsizing) margin and the replace transition.

Appendix: How rare is any non-maintain action?

Movement is rare. 98% of facility-years are pure maintain; only $\sim 2\%$ close anything. The model is identified off that thin closure margin — but with 2.28M facility-years it is still well-measured.

The model's three actions	Facility-years	Share
Maintain	2,237,292	98.07%
Exit (close, no install)	37,967	1.66%
Replace (close + install)	6,183	0.27%

A clean partition of every facility-year. Replacement is the rarest action, 0.27% — closures are too rare to chop into many bins, so we keep three actions.

Appendix: How the model codes each facility-year

The model reads one representative tank per facility and labels each year by what the facility did. Let $c_{it} = 1$ if the facility closes ≥ 1 tank, and n_{it} = new tanks installed that year:

Maintain : $c_{it} = 0$

Exit : $c_{it} = 1, \quad n_{it} = 0, \quad \text{facility ends}$

Replace : $c_{it} = 1, \quad n_{it} \geq 1$

What these labels miss: a facility that closes one tank of several but keeps operating is coded **Maintain**; a pure downsizing (close, no install) is coded **Exit** if it was single-tank, or **Maintain** if multi-tank. There is *no* “shrink’ ’ action.

Next: how often the labels and the underlying events disagree.

Appendix: What facilities do, and how the model bins it

What the facility did	Model bin	% sample
No tank closed	Maintain	97.690%
Full exit (all tanks closed)	Exit	1.682%
Downsizing (closed some, kept op.) [†]	Maintain	0.356%
Replace: same wall (DW→DW, SW→SW)	Replace	0.146%
Replace: SW→DW upgrade [‡]	Replace	0.126%

[†] Downsizing is coded Maintain (no partial-closure action). [‡] The model's Replace assumes every replace is an SW→DW reset — but only ~46% are.

Rows = what facilities actually do; “Model bin” = the 3-action label the estimator assigns. The two daggered rows are where the coding loses information — downsizing and the upgrade/same-wall distinction. Next: how the regime moves each margin.

Appendix: Where the regime bites — FF vs. RB

What the facility did	FF rate	RB rate
No tank closed	97.699%	97.619%
Full exit (all tanks closed)	1.667%	1.793%
Downsizing (closed some, kept op.) [†]	0.383%	0.158%
Replace: same wall (DW→DW, SW→SW)	0.143%	0.174%
Replace: SW→DW upgrade [‡]	0.108%	0.256%

FF/RB rate = share of that regime's facility-years. The RB response concentrates in **full exit** and **SW→DW upgrade** (both rise); downsizing *falls* under RB.

The same rows, split by regime — this is the identifying variation the structural premium and risk coefficients are estimated from.

Appendix: Why one tank is not one facility

The model is a single tank. Each facility is collapsed to one tank on a 32-cell grid (age $\times$ wall $\times$ regime), choosing keep / exit / replace — where “replace” means *reset to one brand-new tank*.

But a facility is a portfolio of tanks, acted on one at a time. Three things follow that a single tank cannot represent:

1. **No partial closure.** Big facilities close one tank of several — the model only has “keep all” or “close all.”
2. **A facility can “get younger.”** Scrap its oldest tank and its average age *falls* — impossible for one aging tank.
3. **“Replace” rarely resets anything.** Replace one tank and the rest remain, so the facility almost never becomes “one new tank.”

Size is a real, fixed dimension the model leaves out — 98–99% of facilities keep the same tank count year to year. Next: the data behind each problem, and how the model could be fixed.

Varies by size →

“Replace” →

Ruled-out moves →

How to fix →

Appendix: How the model could capture this

Three ways to bring the portfolio into the model, in increasing cost:

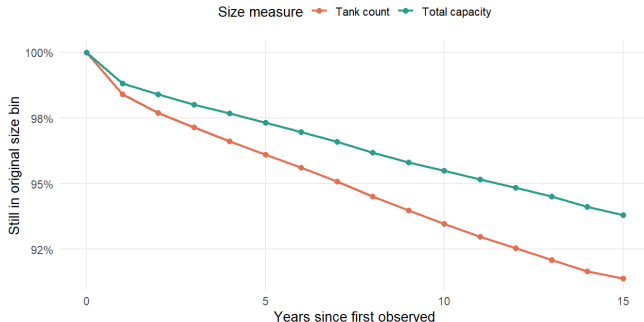
1. **Size as a fixed state dimension.** Add facility size (1 / 2 / 3 / 4+ tanks) to the state, with premiums and hazards re-aggregated by size. Size is 98–99% sticky, so it can enter as a fixed type — no new transition to estimate. *Captures the size-varying choice probabilities; cheapest, and the natural first step.*
2. **A partial-closure action.** Add a fourth choice — close some tanks, keep operating — so downsizing is modeled directly instead of being forced into exit or replace. *Captures the downsizing margin itself.*
3. **A full portfolio model.** Track the facility as a collection of tanks with a joint state. *Most faithful, most complex.*

Planned path: size-as-state first, then the partial-closure action. Both leave the premium/hazard channels — the clean part of the current estimates — intact.

Appendix: Facility size is a near-fixed attribute

Facility size is a near-fixed type, especially by capacity

Among facilities observed at least h years; consolidation preserves capacity, so capacity is stick

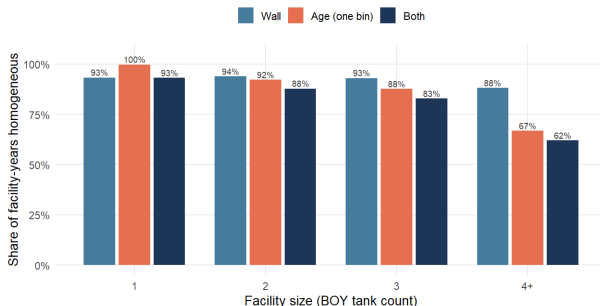


*The fixed-type test is not the one-year stay rate (99%) — size must stay put over a facility's whole spell (mean 16 yrs). It does: ~ 92% of facilities never change their **tank-count** bin and ~ 94% never change their **capacity** bin. Capacity is stickier (consolidation drops a tank but holds capacity), so it is the natural fixed-type size state — entered with no transition to estimate.*

Appendix: Within-facility homogeneity — one tank summarizes the

Within-facility homogeneity falls with size

Share of facility-years whose tanks share one wall / one 5-yr age bin / both. Single-tank are trivial

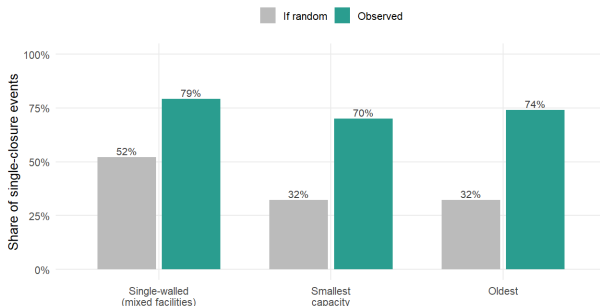


*The other half of “size as a fixed type”: is one representative tank a faithful summary? Across facility-years, 81.5% have all tanks in one wall type and one 5-year age bin — the exact support of the state. The median facility at every size has zero age spread (tanks installed together); heterogeneity is an age-spread phenomenon that emerges only at **4+**-tank facilities (62% homogeneous). So the single representative tank is exact for the bulk of facility-years — the histogram earns its keep only in the large-facility tail*

Appendix: Which tank do they shed? The margin is predictable

The shed tank is the small, single-walled one --- not random

When a multi-tank facility closes one tank, what kind is it? Observed vs a random-tank benchmark

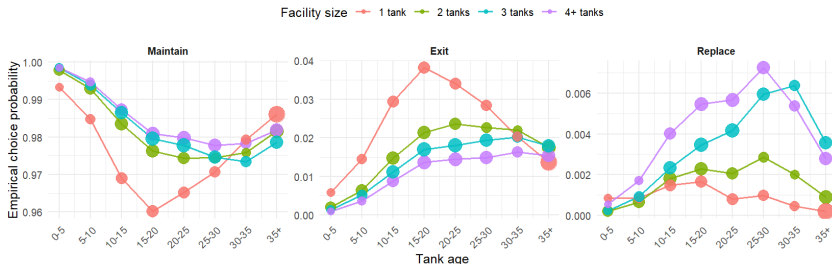


When a multi-tank facility closes one tank, it is not random: it is the **single-walled** tank (79% in mixed facilities vs. 52% by chance), the **smallest** (70% vs. 32%), and more weakly the oldest. So even at large heterogeneous facilities the acted-on tank is predictable — the model can carry the portfolio with a marginal-tank state (the small SW tank) and a capacity-preserving step-down transition, no full histogram needed.

Appendix: Choice probabilities fan out by facility size

Choice probabilities across the state space, by facility size

Faceted by action (free y-scale); point size = cell sample size

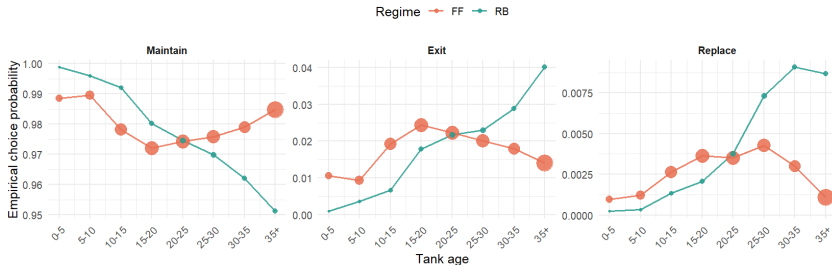


*Empirical choice probabilities by facility size (point size = cell sample size). **Exit** concentrates at single-tank facilities; “**replace**” concentrates at large facilities and young tanks — portfolio churn, not tanks wearing out. The single-tank model has one probability per cell and cannot represent this spread.*

Appendix: Single-walled CCPs across regimes

Single-walled tanks: choice probabilities by regime

Faceted by action (free y-scale); point size = cell sample size

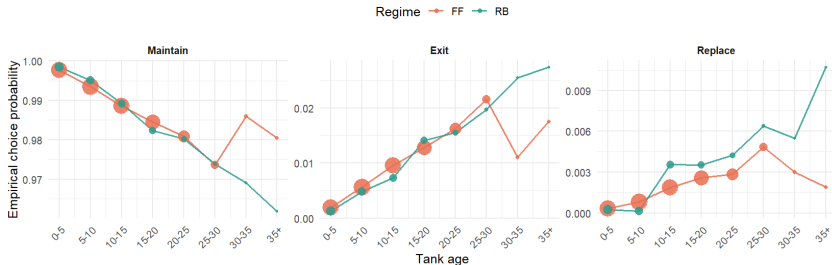


Single-walled tanks, flat-fee vs. risk-based. Under risk-based pricing, exit rises steeply at old ages (middle panel) — the premium channel the structural model captures cleanly.

Appendix: Double-walled CCPs across regimes

Double-walled tanks: choice probabilities by regime

Faceted by action (free y-scale); point size = cell sample size

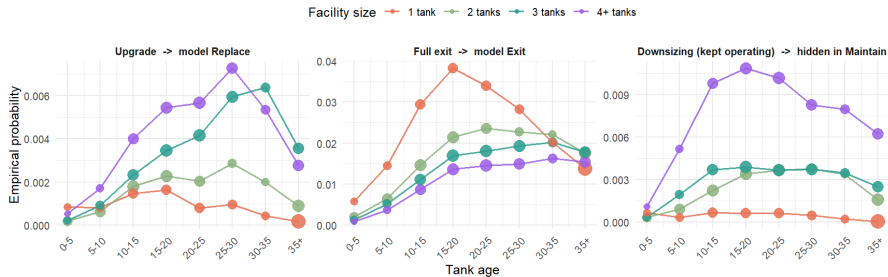


Double-walled tanks, flat-fee vs. risk-based (canonical estimation-panel coding). Both exit and replacement rise with tank age; risk-based pricing steepens the old-age response.

Appendix: Where closures go, by facility size

The three closure events by facility size, and where the model puts them

Empirical probability by facility size; faceted (free y-scale); point size = cell sample size

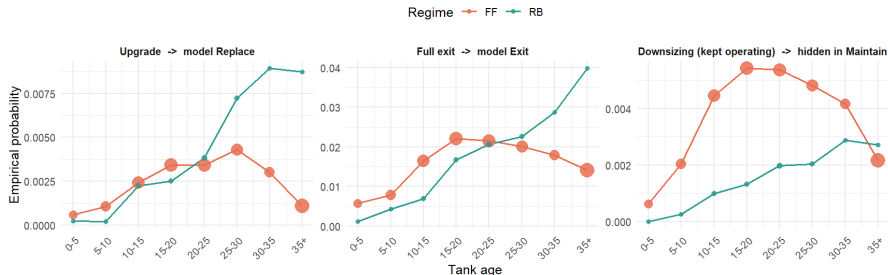


The three closure events by facility size. Upgrades → Replace (left) and full exits → Exit (middle) are coded correctly. **Downsizing** (right; closed a tank but kept operating) is hidden in Maintain — and it rises sharply with facility size, the omitted-state (portfolio) problem. Note the differing y-scales.

Appendix: Where closures go, by regime (FF vs. RB)

The three closure events by regime, and where the model puts them

Under RB the old-tank jump is mostly full exit (captured as Exit); downsizing (hidden in Maintain) is the smaller, uncaptured margin

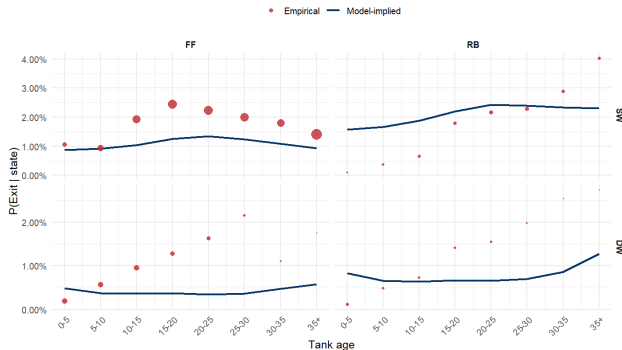


*The same three events, split by regime. Under risk-based pricing the old-tank jump is mostly **full exit** (middle; coded Exit — captured cleanly). **Downsizing** (right; hidden in Maintain) is the smaller, uncaptured margin — and is not even RB-driven. The model captures the dominant RB response; the gap is the partial-downsizing margin.*

Appendix: Model fit — Exit (the margin that matters)

Model fit --- Exit: model-implied vs. empirical

Line = 6p model-implied CCP; points = empirical share (sized by cell N); by wall x regime

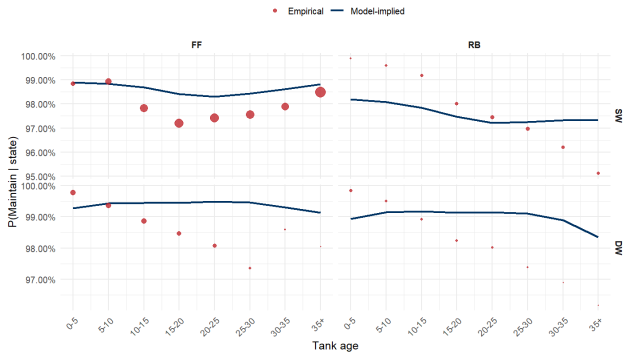


Exit is the dominant non-maintain action (1.7% of years) and the channel the welfare results run through. The model captures its shape — rising with age, higher under RB at old tanks — though it undershoots single-walled flat-fee exit at mid-ages.

Appendix: Model fit — Maintain

Model fit --- Maintain: model-implied vs. empirical

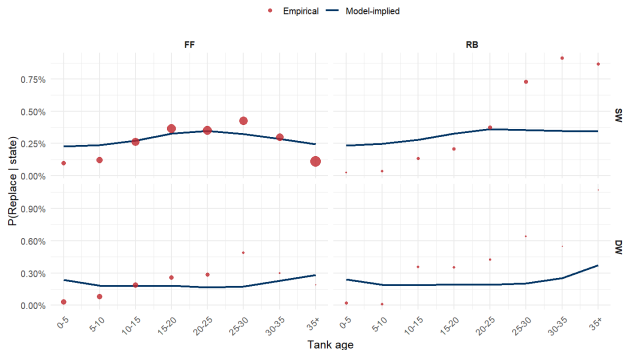
Line = 6p model-implied CCP; points = empirical share (sized by cell N); by wall x regime



Appendix: Model fit — Replace (rare, low-stakes)

Model fit --- Replace: model-implied vs. empirical

Line = 6p model-implied CCP; points = empirical share (sized by cell N); by wall x regime

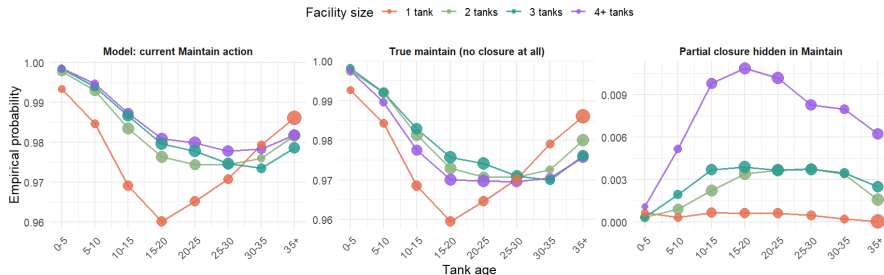


Replace is only 0.27% of years. The model smooths the empirical old-RB bump (e.g. SW-RB reaches $\sim 0.9\%$ at 35+ vs a model line near 0.35%), but because the action is so rare this has little leverage on the estimates — **exit is the margin that matters, and it is captured.**

Appendix: What the model calls “maintain” ’ — hidden partial closures

What the model calls "maintain": true maintain vs. hidden partial closures

Empirical probability by facility size; faceted (free y-scale); point size = cell sample size

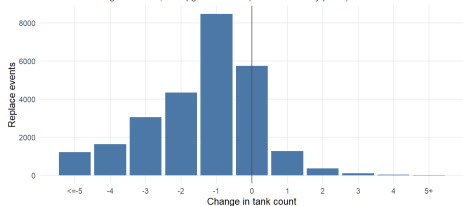


Almost all of “maintain” is genuine (left $\approx$ middle). The sliver that is not is **partial closures** — a facility dropped a tank but kept operating, so it is coded maintain (right). That sliver scales with facility size: $\sim 1.1\%$ per year at 4+-tank facilities vs. $\sim 0.05\%$ at single-tank facilities. The margin the model misses is, mechanically, a **size** dimension.

Appendix: At “replace” events, facilities shed tanks

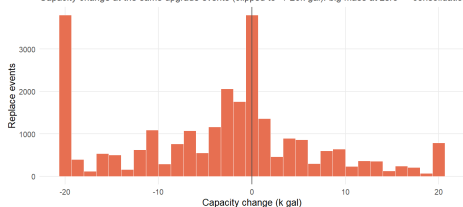
At “replace” events, facilities mostly shed tanks

Tank-count change at the 26,253 upgrade events (canonical facility panel)



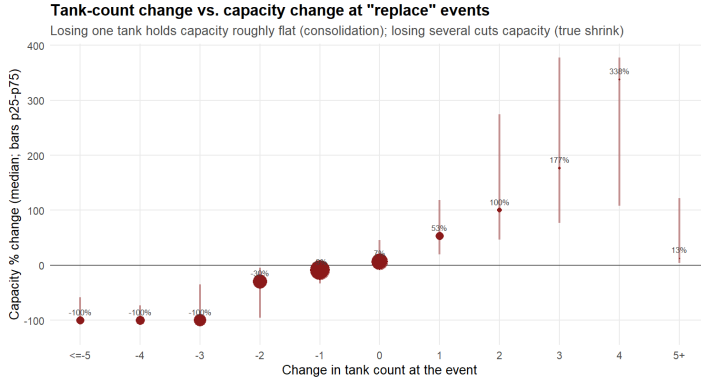
...but total capacity is largely maintained

Capacity change at the same upgrade events (clipped to +/-20k gal): big mass at zero — consolidation



At the 26,253 “replace” events: tank count mostly falls (top), yet total capacity is largely maintained (bottom). So the common changes (0 and −1 tanks) are **consolidations** — fewer,

Appendix: Tank loss and capacity loss move together

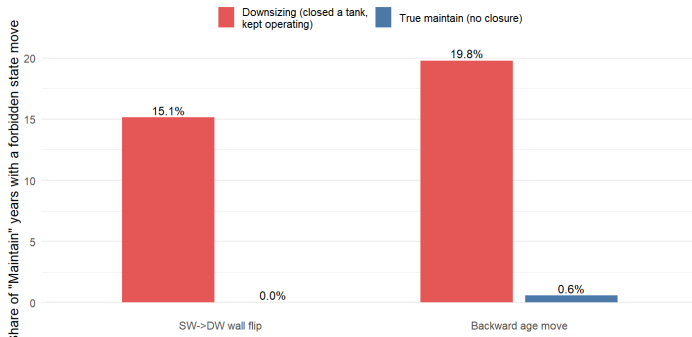


Capacity change vs. tank-count change at "replace" events. Losing one tank holds capacity roughly flat (consolidation — fewer, larger tanks); losing several cuts capacity sharply (true shrinking). Either way the facility is resized, not reset to one new tank.

Appendix: Downsizing corrupts the state, not just the action

The state also breaks: forbidden moves hide inside "Maintain"

Model Maintain keeps wall fixed and ages the tank +1; downsizing years break both rules



*Model Maintain holds wall fixed and ages the tank +1. The downsizing years coded Maintain break both: ~ 15% flip **SW**→**DW** (drop the SW tank, only DW remains — 65% of these flips closed an SW tank) and ~ 20% jump backward in age (drop the oldest tank). True maintains do neither.*

The off-support transitions are not just a mislabeled action — the state itself moves where Maintain forbids.

Appendix: Does the reduced form share this problem?

No — the reduced form is facility-level and portfolio-aware. It splits closures into three separate, well-defined facility outcomes:

- ▶ **Full exit** — the facility disappears; all tanks gone.
- ▶ **Upgrade** — a tank closed *and* a new tank installed the same year.
- ▶ **Shrink** — a tank removed with *no* install; the facility keeps operating, smaller.

A facility can be coded as *both* upgrade and shrink in one year. None of these impose a single-tank reset.

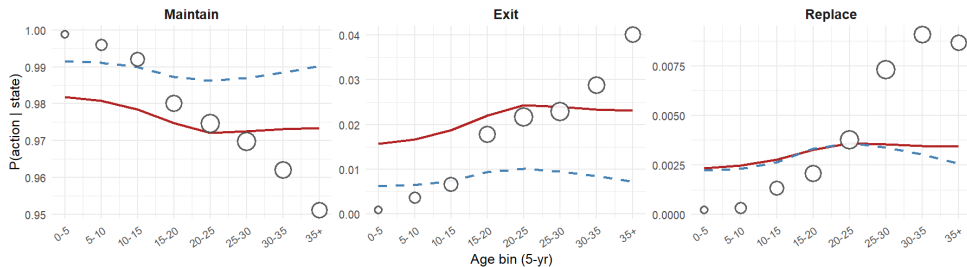
So the reduced-form “composition shifts toward replacement” result is sound. The single-tank reset that misses 98% of replacements is specific to the structural model.

Appendix: TX-on-FF counterfactual — SW action shares

6p+FE model fit + CF overlay — Single-Walled (SW) (TX cells only)

Solid red = baseline (RB) model; dashed blue = CF (FF) model; open circle = TX empirical (size $\propto$ n obs)

Series — Baseline (RB) — CF: TX FF pricing — Empirical (TX data)

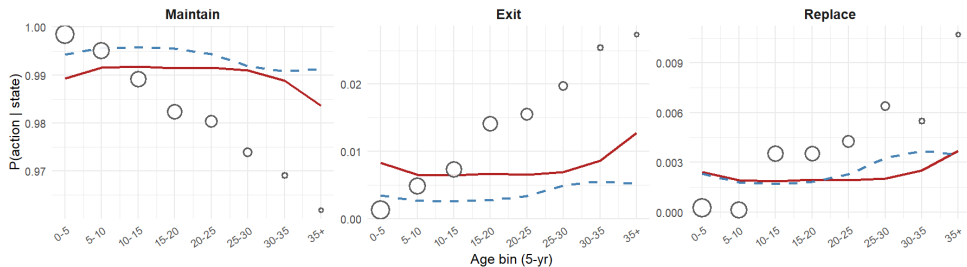


Appendix: TX-on-FF counterfactual — DW action shares

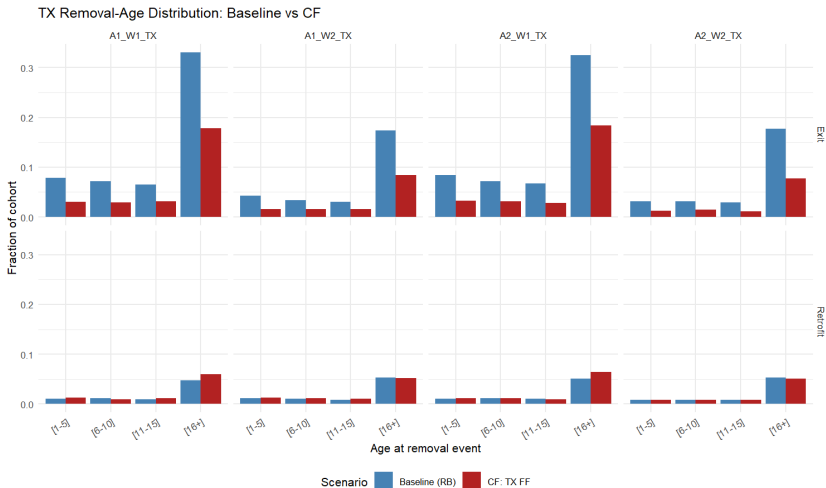
6p+FE model fit + CF overlay — Double-Walled (DW) (TX cells only)

Solid red = baseline (RB) model; dashed blue = CF (FF) model; open circle = TX empirical (size $\propto$ n obs)

Series — Baseline (RB) — CF: TX FF pricing — Empirical (TX data)



Appendix: TX-on-FF counterfactual — removal-age distribution



A dynamic discrete-choice model of tank operations

Scale-incorporation model. A facility is a *portfolio* of tanks, not one representative tank.

- ▶ **State** $s = (n, G, \rho)$: tank counts n over 16 categories (wall $\times$ 8 age bins), capacity quartile G , fixed regime ρ
- ▶ Each period, choose from the menu (k, m) — remove k tanks (where $0 \leq k \leq n$), install m new ones:
 - ▶ $(0, 0)$ **maintain** (keep all tanks)
 - ▶ $(k, 0)$ **downsize** (remove some tanks, $0 < k < n$)
 - ▶ $(k, m \geq 1)$ **replace/consolidate** (remove k , install m newer ones)
 - ▶ $k = n$ **full exit** (exit the portfolio entirely)
- ▶ The choice trades the current flow payoff against the continuation value of the *portfolio*

The single-tank model's keep / exit / replace becomes a per-unit-priced menu over the whole portfolio — recovering the downsizing margin and the size-dependent regime response.

Flow utilities: one payoff per action

$$u_{(k,m)}(s) = \varphi_G + \alpha_g - \gamma_{\text{price}} P(n', \rho) - \gamma_{\text{risk}} \text{OOP}(n') - c^{\text{rem}} k - c^{\text{inst}} m \quad (\text{operate})$$

$$u_X(s) = \kappa_1 N(n) \quad (\text{full exit})$$

- ▶ $P(n, \rho) = \sum_c n_c \bar{p}_c$ (risk-based, empirical cell prices) or $\tau_s N(n)$ (flat fee);
 $\text{OOP}(n) = H(n) \min\{D, L\} = H(n) D$
- ▶ **Estimated** (in green): φ_G operating payoff by size, γ_{price} price / γ_{risk} risk sensitivity, c^{rem} , c^{inst} per-tank removal / install costs, κ_1 exit-size gradient, λ nest
- ▶ α_g state FE — enters the operating payoff only

Two per-unit costs price the whole (k, m) menu — the owner chooses k and m ; the model never predicts them. One model unit = \$10k/yr/facility.

Keep, scrap, or replace: the dynamic stopping problem

$$v_a(s) = u_a(s) + \beta \sum_{s'} F_a(s' | s) V(s'), \quad \beta = 0.9957$$

tank-work nest: $IV(s) = \lambda \log \sum_{(k,m) \neq (0,0)} e^{v_{(k,m)}(s)/\lambda}$

$$V(s) = \log \left(e^{v_{(0,0)}(s)} + e^{IV(s)} + e^{v_X(s)} \right)$$

- ▶ **Maintain** $(0, 0)$ — flow payoff today, keep the portfolio and its option value
- ▶ **Tank-work** — the (k, m) menu (downsize / replace / consolidate), bundled by a shared draw λ : captures the unobserved factors the facility owners know that effects the tank work decision
- ▶ **Exit** X — take $\kappa_1 N(n)$ and leave (absorbing)

From facility-year to (k, m) . How the model codes each action

Let k_{it} = tanks closed and m_{it} = tanks installed in year t . Each facility-year gets exactly one action, by precedence:

Exit X : every tank closed

Replace $(k, m \geq 1)$: $k_{it} \geq 1$, $m_{it} \geq 1$, still operating

Downsize $(k, 0)$: $k_{it} \geq 1$, $m_{it} = 0$, still operating

Maintain $(0, 0)$: $k_{it} = 0$

(Expansion, install without closing $(0, m)$ is not allowed in the action set.)

- ▶ **State** coded from the beginning-of-year portfolio: counts n over 16 (wall $\times$ age) cells, capacity bin G , regime ρ
- ▶ **Which tanks leave** is the rule $r(n, k)$ — oldest single-walled first — *not* a choice; only the counts k , m are chosen
- ▶ The new tank enters double-walled at age bin 1 (regulation)

Downsizing — coded “maintain” or “exit” in the single-tank model — is now its own action; the same panel, re-binned onto the four-action menu

Laws of motion

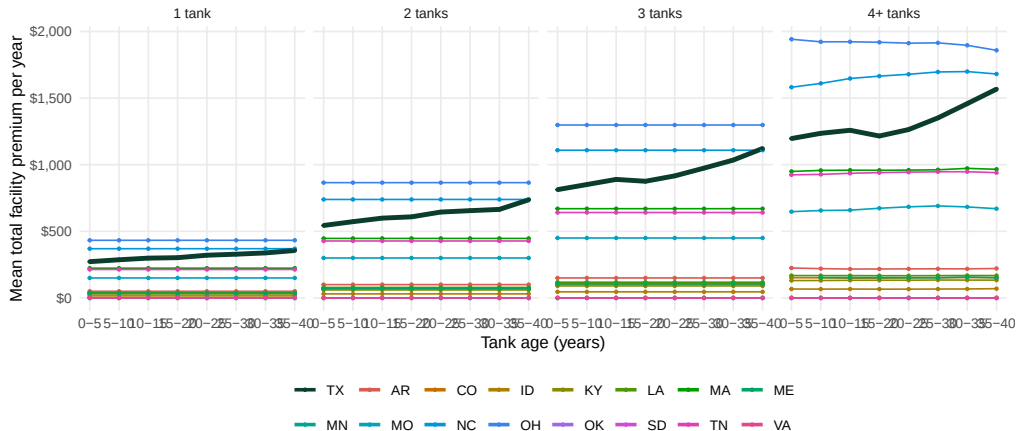
Removal rule $r(n, k)$: the k oldest single-walled tanks leave first; installs enter double-walled at age bin 1; survivors age via Λ ; capacity moves via $\Gamma_G(\cdot \mid G, m-k)$.

Risk-based pricing raises the cost of carrying old single-walled tanks $\Rightarrow$ the owner downsizes or upgrades the risky tank sooner — the margin the single-tank reset could not represent.

A thought experiment. The same tank in two worlds

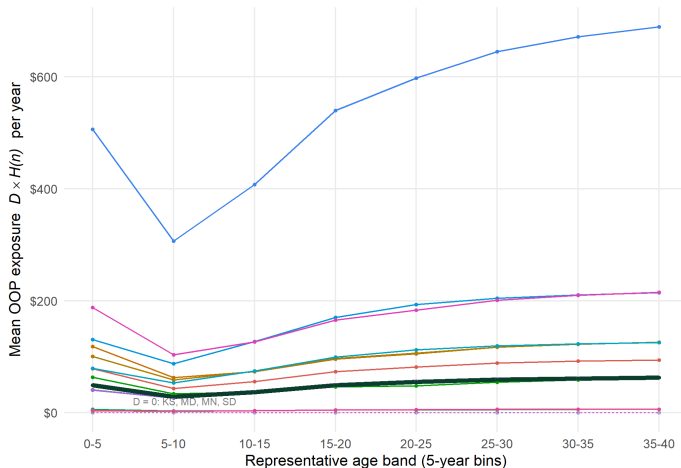
Take one identical facility same leak hazard H and drop it into each regime's contract (premium *and* deductible): [WANT THIS TO BE A VERY SIMPLE MATH EXAMPLE WITH THE TABLE?]

Premium $\times$ age: the variation that identifies γ_{price}



Mean total facility premium by representative age band, one line per state, faceted by size (count held fixed, walls pooled). Texas (risk-based, bold) rises with age; every flat-fee state is flat in age ($P = \tau \cdot N$). The cross-state slope contrast at a given age and size is what pins γ_{price} .

Deductible $\times$ hazard: the variation that identifies γ_{risk}



One point per state: deductible D_s (vertical) against mean facility hazard (horizontal); whiskers span the within-state 10–90% hazard range. Since $\text{OOP} = \gamma_{\text{risk}} H(n) D_s$, the across-state

